RH_CHECKPOINT – Public-Safe Localized Bridge Version

Transport–Spectral Contradiction Program for the Riemann Hypothesis
Closure, Drift, No-Hiding, Gaussian Regularization, and Model Bricks

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Abstract

This document presents a localized transport–spectral checkpoint for the Riemann Hypothesis. It is not an unconditional proof. Its purpose is to isolate a precise conditional contradiction mechanism inside a dipole-regularized residue channel.

The central strategy is to localize the analysis around a single candidate off-critical zero

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

On the closure side, a localized exact window law forces the localized excess energy to vanish:

$$E_{\mathrm{ex}}^{(\rho)}(f) = 0.$$

On the obstruction side, a transported drift law of the form

$$\delta_\rho(u) = \frac{\beta - \frac{1}{2}}{2\pi} u + O(1)$$

would generate a quartic penalty satisfying

$$P_4(\rho; \tau, L) \gtrsim \tau^4 L.$$

If the localized witness bank detects this drift coercively, and if cross/remainder leakage is subcritical, then the same localized channel satisfies

$$E_{\mathrm{ex}}^{(\rho)}(f) > 0,$$

contradicting closure.

The document separates three layers: rigorous algebraic implications, idealized model lemmas, and genuine zeta-side analytic tasks. The appendices prove four supporting model bricks: soft survival of a transported readout, pure-exponential localized witness coercivity, nonstationary cross-term decay, and Gaussian heat-semigroup regularization of singular readouts. A final appendix records the Source–Love Channel as a separate meta-model for contraction, exploration, and stable readout.

Thus the checkpoint does not solve RH. It reduces the remaining work to explicit analytic bottlenecks and states precisely where an unconditional proof would still have to be supplied.

Contents

1 Public Clarification **2**

2 Introduction **3**

3 Why Localization Matters **3**

4 Dependency Ledger: What Depends on What **4**

5 Fourier Preliminaries **4**

6 Dipole-Regularized Residue Channel **6**

7 Localized Witness Bank **6**

8 Localized Exact Window Law **6**

9 Transported Drift and Quartic Obstruction **7**

10 Localized Bridge Package **8**

11 Bank-Subcritical Remainder Criterion **9**

12 Uniform Gaussian Gram Coercivity: Status **10**

13 Master Conditional Contradiction **11**

14 Reality Check: What Is Proved and What Remains Open **12**
14.1 Proved inside this document . . . 13
14.2 Still open on the genuine zeta side . . . 13

15 Outlook and Companion Architecture **14**
15.1 Parity-Channel Bridge . . . 14
15.2 Bressan-type reduction . . . 14
15.3 Readout frames and operator semantics . . . 14

16 Conclusion **15**

A Soft Survival of a Transported Readout **15**
A.1 Soft phase survival above baseline . . . 15

B Pure Exponential Coercivity Model **16**

C Nonstationary Cross-Term Decay **18**

D Gaussian Hammer: Heat-Semigroup Regularization **19**

E Source–Love Channel 2.0: A Separate Meta-Model **21**
E.1 Fixed point via Neumann series . . . 22
E.2 Progress as reduction of epistemic debt . . . 23
E.3 Continuous-time limit . . . 23

1 Public Clarification

Important clarification. The Transport–Spectral Contradiction Program is not presented here as an unconditional proof of the Riemann Hypothesis. It is a conditional reduction isolating exact analytic bottlenecks: transported drift, localized witness coercivity, no-hiding leakage, bank-subcritical remainder, uniform local Gram control, and exact window closure on the genuine zeta residue channel.

Důležité upřesnění. Transportně-spektrální program zde není prezentován jako nepodmíněný důkaz Riemannovy hypotézy. Jde o podmíněnou redukci, která izoluje přesná analytická hrdla: transportovaný drift, lokalizovanou koercitivitu svědecké banky, no-hiding leakage, subkritický zbytek, uniformní lokální Gramovu kontrolu a exact window closure na skutečném zeta-reziduálním kanálu.

Remark 1.1. *This clarification is part of the mathematical content of the checkpoint. It fixes the logical status of the program. The model appendices prove controlled model statements. The RH implication requires transferring the corresponding estimates to the genuine zeta residue channel.*

2 Introduction

The Riemann Hypothesis asserts that every nontrivial zero

$$\rho = \beta + i\gamma$$

of the Riemann zeta function satisfies

$$\beta = \frac{1}{2}.$$

The present framework does not attempt to locate all zeros directly. Instead, it asks whether an off-critical zero would necessarily create a detectable energetic obstruction in a localized readout channel.

The intended contradiction has the form

$$\boxed{\text{localized closure} \Longrightarrow E_{\mathrm{ex}}^{(\rho)}(f) = 0}$$

and

$$\boxed{\text{off-critical transported drift + no-hiding} \Longrightarrow E_{\mathrm{ex}}^{(\rho)}(f) > 0.}$$

Thus the program reduces RH to the verification of a localized exact window law and a localized bridge package for every candidate off-critical channel.

This document is therefore a checkpoint. It is not a final proof. It is a map of the proof obstruction.

3 Why Localization Matters

Global approaches to RH must control the full zeta function at once. The present framework instead asks whether one candidate off-critical zero can be isolated into a local channel whose energy balance becomes contradictory.

Localization provides three advantages:

(i) **Single-channel obstruction.** A candidate zero ρ is treated as an individual transported mode, rather than as an anonymous contribution inside the full global zero set.

(ii) **Explicit interference estimates.** The interaction between the candidate mode and the remainder can be formulated as a localized cross term and attacked by nonstationary-phase estimates.

(iii) **Modular bottlenecks.** The proof decomposes into drift, coercivity, leakage, remainder control, Gram stability, and closure. Each piece can be tested separately.

Localization does not weaken the statement. It sharpens the target.

4 Dependency Ledger: What Depends on What

The document separates four logical layers.

Layer	Content	Status
Definitions	localized bank, excess, window law, drift, penalty	structural
Algebraic implications	closure, bridge inequality, fixed-channel contradiction	proved here
Model bricks	soft survival, pure exponential coercivity, cross-term decay	proved in models
Zeta-side tasks	drift law, coercivity, leakage, Gram bounds, exact window	open

The proof chain is not

$$\text{model lemma} \Longrightarrow \text{RH}.$$

It is

$$\text{model lemma} \Longrightarrow \text{evidence that the analytic target is nonempty.}$$

The actual RH implication requires the corresponding statements on the genuine zeta residue channel.

$$\boxed{\text{Exact Window Law} \Longrightarrow E_{\text{ex}}^{(\rho)}(f) = 0}$$

$$\boxed{\text{Transported Drift} + \text{Witness Coercivity} + \text{No-Hiding Leakage} \Longrightarrow E_{\text{ex}}^{(\rho)}(f) > 0}$$

The contradiction occurs only when both boxes are verified for the same localized zero-channel.

5 Fourier Preliminaries

We use the Fourier transform convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} \, \mathrm{d}x.$$

Let

$$\psi(x) = e^{-\pi x^2}.$$

Lemma 5.1 (Gaussian self-duality)**.** *With the above convention,*

$$\widehat{\psi} = \psi.$$

Proof. Let

$$F(\xi) := \widehat{\psi}(\xi).$$

Differentiating under the integral sign,

$$F'(\xi) = \int_{\mathbb{R}} e^{-\pi x^2} (-2\pi i x) e^{-2\pi i x \xi} \, \mathrm{d}x.$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}x} e^{-\pi x^2} = -2\pi x e^{-\pi x^2},$$

we obtain

$$F'(\xi) = i \int_{\mathbb{R}} \frac{\mathrm{d}}{\mathrm{d}x} (e^{-\pi x^2}) e^{-2\pi i x \xi} \, \mathrm{d}x.$$

Integration by parts gives

$$F'(\xi) = -2\pi \xi F(\xi).$$

Thus

$$F(\xi) = Ce^{-\pi\xi^2}.$$

At $\xi = 0$,

$$C = F(0) = \int_{\mathbb{R}} e^{-\pi x^2}\, \mathrm{d}x = 1.$$

Therefore

$$F(\xi) = e^{-\pi\xi^2}.$$

□

Since the Fourier transform satisfies

$$\mathcal{F}^4 = I$$

on $L^2(\mathbb{R})$, its spectral values lie in

$$\{1, -i, -1, i\}.$$

For each

$$\lambda \in \{1, -i, -1, i\},$$

define the spectral projector

$$P_\lambda := \frac{1}{4} \sum_{m=0}^{3} \lambda^{-m} \mathcal{F}^m.$$

Proposition 5.2 (Fourier spectral projectors)**.** *For every $\lambda \in \{1, -i, -1, i\}$,*

$$P_\lambda^2 = P_\lambda, \qquad \mathcal{F}(P_\lambda f) = \lambda P_\lambda f.$$

In particular,

$$P_{+1} = \frac{1}{4}\left(I + \mathcal{F} + \mathcal{F}^2 + \mathcal{F}^3\right)$$

projects onto the Fourier-fixed self-dual sector.

Proof. The identity $\mathcal{F}^4 = I$ implies that the minimal polynomial of $\mathcal{F}$ divides

$$z^4 - 1 = (z-1)(z+i)(z+1)(z-i).$$

The operators P_λ are the corresponding Lagrange spectral projectors. Directly,

$$\mathcal{F}P_\lambda = \frac{1}{4} \sum_{m=0}^{3} \lambda^{-m} \mathcal{F}^{m+1}.$$

Reindexing the sum modulo 4 and using $\lambda^4 = 1$, one obtains

$$\mathcal{F}P_\lambda = \lambda P_\lambda.$$

Orthogonality and idempotence follow from the finite Fourier orthogonality relation on the cyclic group of order 4:

$$\frac{1}{4} \sum_{m=0}^{3} (\lambda^{-1}\mu)^m = \begin{cases} 1, & \lambda = \mu, \\ 0, & \lambda \neq \mu. \end{cases}$$

Thus $P_\lambda^2 = P_\lambda$. □

Remark 5.3 (Fourier as phase accounting)**.** *In the Hermite basis $\{h_n\}_{n\geq 0}$,*

$$\mathcal{F}h_n = (-i)^n h_n.$$

Thus Fourier evolution is pure phase cycling on Hermite modes:

$$1,\ -i,\ -1,\ i.$$

In the correct basis, Fourier does not mix structure chaotically. It rotates phase mode-by-mode.

6 Dipole-Regularized Residue Channel

Let μ denote the residue distribution arising from a fixed explicit-formula normalization. Let

$$\psi \in \mathcal{S}(\mathbb{R})$$

be a Schwartz test function satisfying the dipole condition

$$\widehat{\psi}(0) = 0.$$

Define the dipole-regularized residue channel

$$f := \psi * \mu.$$

The dipole condition removes the constant component and leaves a fluctuation channel suitable for second-difference witness tests.

For $a > 0$, define the second-difference witness operator

$$(W_a h)(u) := h(u+a) - 2h(u) + h(u-a).$$

Remark 6.1. *The operator W_a annihilates affine functions and detects curvature at scale a. This is why it is suited to measuring drift-induced obstruction.*

7 Localized Witness Bank

Fix a candidate zero

$$\rho = \beta + i\gamma.$$

Let $w_\rho(u) \geq 0$ be a localization window. Let

$$a_j > 0, \qquad j = 0, \ldots, J,$$

be finitely many witness scales, and let $w_j > 0$ be positive bank weights.

Definition 7.1 (Localized witness bank)**.** *The ρ-localized witness-bank energy is*

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) := \sum_{j=0}^{J} w_j^2 \int_\tau^{\tau+L} w_\rho(u)|W_{a_j} f(u)|^2 \, \mathrm{d}u.$$

Definition 7.2 (Localized excess)**.** *Given a baseline density $E_0^{(\rho)} \geq 0$, define the localized excess*

$$E^{(\rho)}_{\text{ex}}(f) := \limsup_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{Q^{(\rho)}_{\text{bank}}(f;\tau,L) - E_0^{(\rho)} L}{L}.$$

8 Localized Exact Window Law

Definition 8.1 (Localized exact window law)**.** *We say that the ρ-localized exact window law holds if there exist a bounded function*

$$J_f^{(\rho)} : \mathbb{R} \to \mathbb{R}$$

and a remainder $R^{(\rho)}(\tau,L)$ such that

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) = E_0^{(\rho)} L + J_f^{(\rho)}(\tau+L) - J_f^{(\rho)}(\tau) + R^{(\rho)}(\tau,L),$$

where

$$\sup_{\tau\in\mathbb{R}} |J_f^{(\rho)}(\tau)| < \infty,$$

and

$$\lim_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{|R^{(\rho)}(\tau,L)|}{L} = 0.$$

Theorem 8.2 (Localized closure)**.** *If the localized exact window law holds, then*

$$E_{\mathrm{ex}}^{(\rho)}(f) = 0.$$

Proof. Divide the exact window identity by L:

$$\frac{Q_{\mathrm{bank}}^{(\rho)}(f;\tau,L) - E_0^{(\rho)}L}{L} = \frac{J_f^{(\rho)}(\tau+L) - J_f^{(\rho)}(\tau)}{L} + \frac{R^{(\rho)}(\tau,L)}{L}.$$

Taking the supremum over τ,

$$\sup_\tau \left| \frac{Q_{\mathrm{bank}}^{(\rho)}(f;\tau,L) - E_0^{(\rho)}L}{L} \right| \le \frac{2\|J_f^{(\rho)}\|_\infty}{L} + \sup_\tau \frac{|R^{(\rho)}(\tau,L)|}{L}.$$

Both terms tend to 0 as $L \to \infty$. Hence

$$E_{\mathrm{ex}}^{(\rho)}(f) = 0.$$

□

9 Transported Drift and Quartic Obstruction

Let

$$\rho = \beta + i\gamma$$

be a candidate off-critical zero.

Definition 9.1 (Transported drift law)**.** *We say that the transported drift law holds for ρ if*

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad a_\rho := \frac{\beta - \frac{1}{2}}{2\pi},$$

where r_ρ is bounded.

Definition 9.2 (Quartic penalty)**.** *Define the quartic penalty by*

$$P_4(\rho;\tau,L) := \int_\tau^{\tau+L} |\delta_\rho(u)|^4 \, \mathrm{d}u.$$

Lemma 9.3 (Quartic lower bound)**.** *Assume the transported drift law and $a_\rho \neq 0$. Then for sufficiently large τ and all $L \ge 1$,*

$$P_4(\rho;\tau,L) \ge c_\rho \tau^4 L$$

for some constant $c_\rho > 0$.

Proof. Write

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad |r_\rho(u)| \le M.$$

For u sufficiently large,

$$|\delta_\rho(u)| \ge |a_\rho| u - M \ge \frac{|a_\rho|}{2} u.$$

Thus for $u \in [\tau, \tau+L]$ and τ sufficiently large,

$$|\delta_\rho(u)|^4 \ge \frac{|a_\rho|^4}{16} u^4 \ge \frac{|a_\rho|^4}{16} \tau^4.$$

Integrating over an interval of length L gives

$$P_4(\rho;\tau,L) \ge \frac{|a_\rho|^4}{16} \tau^4 L.$$

□

Remark 9.4. *The exponent 4 is critical for later comparison. If a remainder contribution grows like*

$$Q_{\mathrm{rest}} \lesssim \tau^{2m_0} L$$

with $m_0 < 2$, then $2m_0 < 4$, and therefore

$$\frac{Q_{\mathrm{rest}}}{P_4} \to 0 \qquad (\tau \to \infty).$$

10 Localized Bridge Package

Assume a single-zero splitting

$$f = f_\rho + f_{\text{rest}}.$$

Then

$$W_{a_j} f = W_{a_j} f_\rho + W_{a_j} f_{\text{rest}}.$$

Accordingly,

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) = Q^{(\rho)}_{\rho}(\tau,L) + Q^{(\rho)}_{\text{cross}}(\tau,L) + Q^{(\rho)}_{\text{rest}}(\tau,L),$$

where

$$Q^{(\rho)}_{\rho}(\tau,L) := \sum_{j=0}^{J} w_j^2 \int_\tau^{\tau+L} w_\rho(u)|W_{a_j} f_\rho(u)|^2 \, \mathrm{d}u,$$

$$Q^{(\rho)}_{\text{cross}}(\tau,L) := 2\sum_{j=0}^{J} w_j^2 \operatorname{Re} \int_\tau^{\tau+L} w_\rho(u) W_{a_j} f_\rho(u) \overline{W_{a_j} f_{\text{rest}}(u)} \, \mathrm{d}u,$$

and

$$Q^{(\rho)}_{\text{rest}}(\tau,L) := \sum_{j=0}^{J} w_j^2 \int_\tau^{\tau+L} w_\rho(u)|W_{a_j} f_{\text{rest}}(u)|^2 \, \mathrm{d}u.$$

Definition 10.1 (Localized witness coercivity)**.** *The localized bank is coercive on f_ρ if there exist $\kappa > 0$ and $C_0 \geq 0$ such that*

$$Q^{(\rho)}_{\rho}(\tau,L) \geq \kappa P_4(\rho;\tau,L) - C_0 L$$

for sufficiently large τ and all $L \geq 1$.

Definition 10.2 (Localized no-hiding leakage)**.** *The localized leakage is subcritical if there exist $C_1 \geq 0$ and $\eta(\tau) \to 0$ such that*

$$Q^{(\rho)}_{\text{cross}}(\tau,L) + Q^{(\rho)}_{\text{rest}}(\tau,L) \geq -\eta(\tau) P_4(\rho;\tau,L) - C_1 L$$

for sufficiently large τ and all $L \geq 1$.

Proposition 10.3 (Localized bridge inequality)**.** *If localized witness coercivity and localized no-hiding leakage hold, then*

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) \geq (\kappa - \eta(\tau)) P_4(\rho;\tau,L) - (C_0 + C_1)L.$$

In particular, for sufficiently large τ,

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) \geq \frac{\kappa}{2} P_4(\rho;\tau,L) - (C_0 + C_1)L.$$

Proof. By the decomposition,

$$Q^{(\rho)}_{\text{bank}} = Q^{(\rho)}_{\rho} + Q^{(\rho)}_{\text{cross}} + Q^{(\rho)}_{\text{rest}}.$$

Using coercivity and no-hiding,

$$Q^{(\rho)}_{\text{bank}} \geq \kappa P_4 - C_0 L - \eta(\tau) P_4 - C_1 L.$$

Thus

$$Q^{(\rho)}_{\text{bank}} \geq (\kappa - \eta(\tau)) P_4 - (C_0 + C_1)L.$$

Since $\eta(\tau) \to 0$, for sufficiently large τ we have

$$\eta(\tau) \leq \frac{\kappa}{2}.$$

The second inequality follows. □

11 Bank-Subcritical Remainder Criterion

The following criterion explains why the exponent $m_0 < 2$ is the natural subcritical threshold. The clean form is integral rather than pointwise in τ, because the comparison must remain stable for long windows.

Lemma 11.1 (Second-difference bound)**.** *Let $h \in C^2([\tau-a, \tau+L+a])$. Then for every $u \in [\tau, \tau+L]$,*

$$|W_a h(u)| \le a^2 \sup_{v\in[u-a,u+a]} |h''(v)|.$$

Proof. By Taylor's theorem with integral remainder,

$$h(u+a) = h(u) + ah'(u) + \int_0^a (a-s)h''(u+s)\ \mathrm{d}s,$$

and

$$h(u-a) = h(u) - ah'(u) + \int_0^a (a-s)h''(u-s)\ \mathrm{d}s.$$

Adding these two expansions and subtracting $2h(u)$, the linear terms cancel:

$$W_a h(u) = \int_0^a (a-s)\big(h''(u+s) + h''(u-s)\big)\ \mathrm{d}s.$$

Therefore

$$|W_a h(u)| \le 2\int_0^a (a-s)\ \mathrm{d}s \sup_{v\in[u-a,u+a]} |h''(v)| = a^2 \sup_{v\in[u-a,u+a]} |h''(v)|.$$

□

Proposition 11.2 (Integral bank-subcritical remainder criterion)**.** *Assume that for sufficiently large u,*

$$|f''_{\mathrm{rest}}(u)| \le C(1+u)^{m_0}$$

with

$$m_0 < 2.$$

Assume also that the localization window is bounded:

$$0 \le w_\rho(u) \le M_\rho.$$

Then, for fixed finite bank scales,

$$Q^{(\rho)}_{\mathrm{rest}}(\tau, L) \le C' \int_\tau^{\tau+L} (1+u)^{2m_0}\ \mathrm{d}u.$$

If the quartic obstruction satisfies the integral lower bound

$$P_4(\rho;\tau,L) \ge c\int_\tau^{\tau+L} u^4\ \mathrm{d}u - C_0 \int_\tau^{\tau+L} u^3\ \mathrm{d}u,$$

then, as the transported window moves to large scale,

$$\frac{Q^{(\rho)}_{\mathrm{rest}}(\tau, L)}{P_4(\rho;\tau,L)} \to 0.$$

In particular,

$$Q^{(\rho)}_{\mathrm{rest}} = o(P_4).$$

Proof. By the second-difference bound, for each fixed a_j,

$$|W_{a_j} f_{\mathrm{rest}}(u)| \le C_j(1+u)^{m_0}.$$

Therefore

$$|W_{a_j} f_{\mathrm{rest}}(u)|^2 \le C_j'(1+u)^{2m_0}.$$

Multiplying by the bounded localization window, summing over finitely many j, and integrating,

$$Q_{\mathrm{rest}}^{(\rho)}(\tau, L) \le C' \int_\tau^{\tau+L} (1+u)^{2m_0}\, \mathrm{d}u.$$

Since $2m_0 < 4$, the numerator grows with strictly lower polynomial order than the quartic integral

$$\int_\tau^{\tau+L} u^4\, \mathrm{d}u.$$

More explicitly, if the transported window moves to large scale, then

$$\frac{\int_\tau^{\tau+L}(1+u)^{2m_0}\, \mathrm{d}u}{\int_\tau^{\tau+L} u^4\, \mathrm{d}u} \to 0.$$

The lower-order correction

$$\int_\tau^{\tau+L} u^3\, \mathrm{d}u$$

is also subcritical relative to the u^4-term at large scale. Hence

$$\frac{Q_{\mathrm{rest}}^{(\rho)}(\tau, L)}{P_4(\rho; \tau, L)} \to 0.$$

□

Remark 11.3. *This version avoids relying on the informal condition $u \asymp \tau$. The comparison is made directly between the integrated remainder growth and the integrated quartic obstruction. The strict inequality*

$$2m_0 < 4$$

is the exact reason why $m_0 < 2$ is the critical threshold.

12 Uniform Gaussian Gram Coercivity: Status

The no-hiding mechanism also suggests a local Gram-stability requirement. This section records the analytic target without claiming it is solved on the genuine zeta side.

Let a local cluster of ordinates be written as

$$\Gamma = \{\gamma_1, \dots, \gamma_N\}.$$

For a fixed Gaussian parameter $\alpha > 0$, define the Gaussian Gram matrix

$$K_{jk} := \exp(-\alpha(\gamma_j - \gamma_k)^2).$$

Lemma 12.1 (Finite-cluster positivity)**.** *If the ordinates $\gamma_1, \dots, \gamma_N$ are distinct, then K is positive definite.*

Proof. The Gaussian kernel is positive definite. More explicitly, by the Fourier representation of the Gaussian,

$$e^{-\alpha(x-y)^2} = c_\alpha \int_{\mathbb{R}} e^{itx} e^{-ity} e^{-t^2/(4\alpha)}\, \mathrm{d}t$$

for a positive constant $c_\alpha > 0$. Hence, for any $c = (c_1, \ldots, c_N) \in \mathbb{C}^N$,

$$\sum_{j,k=1}^{N} c_j \overline{c_k} K_{jk} = c_\alpha \int_{\mathbb{R}} \left| \sum_{j=1}^{N} c_j e^{it\gamma_j} \right|^2 e^{-t^2/(4\alpha)}\, \mathrm{d}t.$$

If this integral is zero, then the exponential polynomial

$$\sum_{j=1}^{N} c_j e^{it\gamma_j}$$

vanishes for all real t. Since the γ_j are distinct, the exponentials are linearly independent, hence all $c_j = 0$. Therefore K is positive definite. □

Open Problem 12.2 (Height-uniform local Gaussian Gram coercivity)**.** *After unfolding local zeta-zero clusters to unit mean spacing, prove a height-uniform lower bound*

$$\lambda_{\min}(K) \geq c(\alpha, L) > 0$$

for all sufficiently high local clusters inside a fixed unfolded window of size L.

Remark 12.3. *Finite-cluster positivity is not enough for the RH bridge. The needed statement is uniform coercivity as height tends to infinity. Diagonal dominance can fail for moderately dense clusters, because nearby ordinates make off-diagonal Gaussian entries large. Therefore the desired estimate cannot rely only on a naive pairwise separation argument; it likely requires global structure of the unfolded zero configuration.*

13 Master Conditional Contradiction

Theorem 13.1 (Fixed-channel contradiction)**.** *Fix a candidate zero*

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

Assume that the following four statements hold for the same localized channel associated with ρ:

(1) *the transported drift law holds:*

$$\delta_\rho(u) = \frac{\beta - \frac{1}{2}}{2\pi} u + O(1);$$

(2) *the localized witness coercivity estimate holds:*

$$Q_\rho^{(\rho)}(\tau, L) \geq \kappa P_4(\rho; \tau, L) - C_0 L;$$

(3) *the localized no-hiding leakage estimate holds:*

$$Q_{\mathrm{cross}}^{(\rho)}(\tau, L) + Q_{\mathrm{rest}}^{(\rho)}(\tau, L) \geq -\eta(\tau) P_4(\rho; \tau, L) - C_1 L, \qquad \eta(\tau) \to 0;$$

(4) *the localized exact window law holds:*

$$Q_{\mathrm{bank}}^{(\rho)}(f; \tau, L) = E_0^{(\rho)} L + J_f^{(\rho)}(\tau + L) - J_f^{(\rho)}(\tau) + R^{(\rho)}(\tau, L),$$

with $J_f^{(\rho)}$ bounded and

$$\lim_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{|R^{(\rho)}(\tau, L)|}{L} = 0.$$

Then a contradiction follows. Therefore no off-critical zero can satisfy these four statements simultaneously.

Proof. Since

$$\beta \neq \frac{1}{2},$$

the coefficient

$$a_\rho = \frac{\beta - \frac{1}{2}}{2\pi}$$

is nonzero. By the transported drift law and the quartic lower bound, for sufficiently large τ,

$$P_4(\rho; \tau, L) \geq c_\rho \tau^4 L.$$

By witness coercivity and no-hiding leakage,

$$Q^{(\rho)}_{\text{bank}}(f; \tau, L) \geq (\kappa - \eta(\tau)) P_4(\rho; \tau, L) - (C_0 + C_1)L.$$

For sufficiently large τ, $\eta(\tau) \leq \kappa/2$, hence

$$Q^{(\rho)}_{\text{bank}}(f; \tau, L) \geq \frac{\kappa}{2} P_4(\rho; \tau, L) - (C_0 + C_1)L.$$

Using the quartic lower bound,

$$Q^{(\rho)}_{\text{bank}}(f; \tau, L) - E^{(\rho)}_0 L \geq c' \tau^4 L - C'' L.$$

For large τ, the right-hand side is positive after division by L. Hence

$$E^{(\rho)}_{\text{ex}}(f) > 0.$$

On the other hand, the localized exact window law implies

$$E^{(\rho)}_{\text{ex}}(f) = 0.$$

This is a contradiction. □

Corollary 13.2 (Conditional RH)**.** *Assume that for every off-critical candidate zero*

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2},$$

the four hypotheses of the fixed-channel contradiction theorem hold. Then every nontrivial zero of $\zeta(s)$ lies on the critical line:

$$\text{Re}(\rho) = \frac{1}{2}.$$

Proof. If an off-critical zero existed, the fixed-channel contradiction theorem would apply to its localized channel. This is impossible. □

14 Reality Check: What Is Proved and What Remains Open

The present document should be read as a conditional checkpoint. It contains several rigorous components, but the genuine zeta-side bridge is not yet complete.

14.1 Proved inside this document

The following implications are proved within the stated framework:

1. The localized exact window law implies
$$E_{\mathrm{ex}}^{(\rho)}(f) = 0.$$
2. The transported drift law implies the quartic lower bound
$$P_4(\rho;\tau,L) \geq c_\rho \tau^4 L$$
for sufficiently large τ.
3. Localized witness coercivity plus localized no-hiding leakage implies the bridge lower bound
$$Q_{\mathrm{bank}}^{(\rho)}(f;\tau,L) \geq (\kappa - \eta(\tau))P_4(\rho;\tau,L) - O(L).$$
4. A C^2 remainder bound with exponent $m_0 < 2$ is bank-subcritical.
5. Finite Gaussian Gram matrices of distinct ordinates are positive definite.
6. In the pure-exponential model with a plateau window, localized witness coercivity holds.
7. Under nonstationary phase assumptions, the cross term is quartic-subcritical:
$$Q_{\mathrm{cross}}^{(\rho)} = o(P_4).$$
8. Gaussian heat-semigroup regularization sends tempered singular readouts to smooth effective profiles and suppresses high frequencies exponentially.

14.2 Still open on the genuine zeta side

The following analytic tasks remain open:

1. **Transported drift law.** Derive
$$\delta_\rho(u) = \frac{\beta - \frac{1}{2}}{2\pi} u + O(1)$$
from a concrete transport normalization of the genuine zeta residue channel.
2. **Localized witness coercivity.** Upgrade the pure-exponential model theorem to the actual transported zero-channel.
3. **Localized no-hiding leakage.** Prove
$$Q_{\mathrm{cross}}^{(\rho)}(\tau,L) + Q_{\mathrm{rest}}^{(\rho)}(\tau,L) \geq -\eta(\tau)P_4(\rho;\tau,L) - C_1 L, \qquad \eta(\tau) \to 0.$$
4. **Bank-subcritical remainder.** Establish residue-side regularity strong enough to imply
$$Q_{\mathrm{rest}}^{(\rho)} = o(P_4).$$
5. **Uniform Gaussian Gram coercivity.** Prove height-uniform lower bounds for local Gaussian Gram matrices associated with actual zeta-zero clusters.
6. **Exact window law on the genuine residue channel.** Verify the localized exact window law for the concrete dipole-regularized channel arising from the explicit formula.

7. **Model-to-zeta bridge.** Transfer the model estimates in the appendices to the explicit-formula residue channel.

Remark 14.1 (Model bricks versus zeta-side proof)**.** *The appendices prove that the localized bridge mechanisms are mathematically nonempty in controlled model regimes. They do not, by themselves, prove the corresponding estimates for the genuine zeta residue channel. The unconditional RH proof would require exactly that transfer.*

Architecture is not proof.

But a good architecture tells us exactly where the proof must happen.

15 Outlook and Companion Architecture

This section records several companion directions that are useful for interpretation and future work, but are not used as proof components in the fixed-channel contradiction theorem.

15.1 Parity-Channel Bridge

A parity-channel bridge would attempt to isolate odd or antisymmetric contributions that survive dipole regularization. In such a setting, an odd-sector projector P_{odd} could serve as a filter for components that carry phase-sensitive drift.

At the current checkpoint level, no theorem in the main proof depends on this bridge. It is an outlook direction for sharpening the no-hiding mechanism and for separating symmetric background from asymmetric transported signal.

15.2 Bressan-type reduction

A Bressan-type reduction, if formulated in this framework, would serve as a compression or transport simplification layer for handling complex channel interactions. In the present document it is not used as an analytic assumption and not used in any proof. It belongs to a future companion architecture.

15.3 Readout frames and operator semantics

The readout-frame principle can be stated as follows: a symbolic expression does not determine a numerical or geometric value until its representation, operator semantics, and projection frame are fixed.

In schematic notation,

$$\mathrm{Val}(E; R, \Sigma, \Theta)$$

depends not only on the expression E, but also on:

- R: the representation space;
- Σ: the semantics of the operators involved;
- Θ: the readout parameters, including projection, phase, and normalization.

This principle is conceptually aligned with the localized RH program: the zeta residue channel must be read through a declared transport, regularization, and witness bank. However, this semantic principle is not itself a proof of any zeta-side estimate.

Remark 15.1. *Parity-channel ideas, Bressan-type reductions, and readout semantics may help guide future formulations, but the proof chain in this document remains exactly the fixed-channel chain: drift, coercivity, no-hiding leakage, exact window, contradiction.*

16 Conclusion

The localized transport–spectral program reduces the Riemann Hypothesis to a precise conditional contradiction mechanism.

The closure side says:

$$E_{\mathrm{ex}}^{(\rho)}(f) = 0.$$

The obstruction side says that an off-critical transported drift should force:

$$E_{\mathrm{ex}}^{(\rho)}(f) > 0.$$

The contradiction becomes valid once the localized bridge package is proved on the genuine zeta residue channel.

At this checkpoint level, the main contribution is not an unconditional proof, but a clean separation of the problem into explicit analytic tasks. The supporting appendices show that the mechanism is nonempty: coercivity, cross-term decay, soft readout survival, Gaussian regularization, and finite-cluster Gram positivity all hold in mathematically controlled regimes.

The remaining work is difficult, but it is now localized, modular, and explicit.

Closure kills leakage.
Obstruction creates drift.
No-Hiding is the lock.
Localization makes the key visible.

A Soft Survival of a Transported Readout

This appendix records a structural compatibility lemma. Soft regularization does not create a transported drift, but it can prevent a pre-existing transported signal from being annihilated by a local hard readout collapse.

A.1 Soft phase survival above baseline

Lemma A.1 (Soft phase survival above baseline)**.** *Let $U \subset \mathbb{R}$ be a neighborhood of a local singular point, let $\varepsilon > 0$, and define*

$$r_\varepsilon(\xi) := \sqrt{\xi^2 + \varepsilon^2}.$$

Assume that the effective readout admits an expansion

$$\Pi_{\eta,\varepsilon}(\xi) = \Pi_0 + \eta W(r_\varepsilon(\xi)) + R_{\eta,\varepsilon}(\xi), \qquad \xi \in U,$$

where:

(i) $\Pi_0 \in \mathbb{R}$ *is the baseline readout level;*

(ii) $W : [0,\infty) \to [0,\infty)$ *is continuous and nonnegative;*

(iii) there exists $C > 0$ such that

$$|R_{\eta,\varepsilon}(\xi)| \le C\eta^2$$

for all $\xi \in U$ and sufficiently small $\eta > 0$;

(iv) there exists $w_* > 0$ such that

$$W(r_\varepsilon(\xi)) \ge w_*$$

for all $\xi \in U$.

Then there exists $\eta_0 > 0$ such that for all $0 < \eta < \eta_0$,

$$\Pi_{\eta,\varepsilon}(\xi) \geq \Pi_0 + \frac{1}{2}\eta w_* > \Pi_0$$

for all $\xi \in U$. If $\Pi_0 = 0$, then

$$\Pi_{\eta,\varepsilon}(\xi) > 0.$$

Proof. By assumption,

$$\Pi_{\eta,\varepsilon}(\xi) = \Pi_0 + \eta W(r_\varepsilon(\xi)) + R_{\eta,\varepsilon}(\xi).$$

Using $W(r_\varepsilon(\xi)) \geq w_*$ and $|R_{\eta,\varepsilon}| \leq C\eta^2$,

$$\Pi_{\eta,\varepsilon}(\xi) \geq \Pi_0 + \eta w_* - C\eta^2.$$

Choose $\eta_0 > 0$ so that

$$C\eta^2 \leq \frac{1}{2}\eta w_*$$

whenever $0 < \eta < \eta_0$. Then

$$\Pi_{\eta,\varepsilon}(\xi) \geq \Pi_0 + \frac{1}{2}\eta w_*.$$

If $\Pi_0 = 0$, this is strictly positive. □

Corollary A.2 (Compatibility with transported drift)**.** *Let f_ρ be a localized transported mode associated with an off-critical zero*

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2},$$

and suppose that in a transported frame its effective phase variable satisfies

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi},$$

with r_ρ bounded.

If the local readout near a hard singularity satisfies the soft expansion in the preceding lemma, then the local loss of hard sign-readout does not force annihilation of the transported signal in the soft channel.

Proof. Apply the previous lemma to the local readout of the transported channel. □

Remark A.3. *This result does not prove the transported drift law for the genuine zeta function. It proves only the following stability statement:*

if a transported drift is already present, soft regularization can keep it visible through a local hard readout collapse.

B Pure Exponential Coercivity Model

This appendix proves localized witness coercivity in an exact pure-exponential transport model with a plateau localization window.

Proposition B.1 (Localized witness coercivity: pure exponential model)**.** *Let*

$$\rho = \beta + i\gamma, \qquad 0 < \beta < 1, \qquad \beta \neq \frac{1}{2}.$$

Define

$$g_\rho(u) := A_\rho e^{\rho u}, \qquad A_\rho := \exp\left(\frac{\rho^2}{4\pi}\right).$$

Assume the transported drift law

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad a_\rho := \frac{\beta - \frac{1}{2}}{2\pi},$$

with r_ρ bounded.

Let $w_{\rho,\tau,L}$ be a plateau window satisfying

$$0 < m \le w_{\rho,\tau,L}(u) \le M < \infty$$

for all $u \in [\tau, \tau + L]$. Then there exist constants

$$\kappa > 0, \qquad C_0 \ge 0, \qquad \tau_0 > 0$$

such that for all $\tau \ge \tau_0$ and all $L \ge 1$,

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) \ge \kappa P_4(\rho; \tau, L) - C_0 L.$$

Proof. Since g_ρ is a pure exponential,

$$W_{a_j} g_\rho(u) = g_\rho(u + a_j) - 2g_\rho(u) + g_\rho(u - a_j) = d_{\rho,a_j} g_\rho(u),$$

where

$$d_{\rho,a_j} := e^{\rho a_j} - 2 + e^{-\rho a_j}.$$

Hence

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) = |A_\rho|^2 \sum_{j=0}^{J} w_j^2 |d_{\rho,a_j}|^2 \int_\tau^{\tau+L} w_{\rho,\tau,L}(u) e^{2\beta u} \, \mathrm{d}u.$$

Set

$$S_\rho := \sum_{j=0}^{J} w_j^2 |d_{\rho,a_j}|^2.$$

We claim that $S_\rho > 0$. Indeed, if $d_{\rho,a_j} = 0$, then

$$e^{\rho a_j} + e^{-\rho a_j} = 2,$$

or

$$\cosh(\rho a_j) = 1.$$

But the solutions of $\cosh z = 1$ are

$$z = 2\pi i k, \qquad k \in \mathbb{Z},$$

so necessarily $\Re z = 0$. This contradicts

$$\Re(\rho a_j) = \beta a_j > 0.$$

Thus $d_{\rho,a_j} \ne 0$ and $S_\rho > 0$.

Since $w_{\rho,\tau,L} \ge m$ on $[\tau, \tau + L]$,

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) \ge |A_\rho|^2 S_\rho m \int_\tau^{\tau+L} e^{2\beta u} \, \mathrm{d}u.$$

Therefore

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) \ge c_1 \left(e^{2\beta(\tau+L)} - e^{2\beta\tau} \right),$$

where

$$c_1 := |A_\rho|^2 S_\rho \frac{m}{2\beta} > 0.$$

Since r_ρ is bounded,

$$|\delta_\rho(u)| \le |a_\rho| u + C$$

for some $C > 0$. Hence

$$P_4(\rho; \tau, L) \le C' \left((\tau + L)^5 - \tau^5 + L \right).$$

For $L \ge 1$,

$$e^{2\beta(\tau+L)} - e^{2\beta\tau} = e^{2\beta(\tau+L)}(1 - e^{-2\beta L}) \ge (1 - e^{-2\beta}) e^{2\beta(\tau+L)}.$$

Exponential growth dominates every polynomial uniformly for large $\tau + L$. Hence, after choosing τ_0 sufficiently large and $\kappa > 0$ sufficiently small,

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) \ge \kappa P_4(\rho; \tau, L) - C_0 L.$$

□

Remark B.2. *This is a rigorous model theorem. It does not yet apply to Gaussian-modulated packets or to the genuine zeta residue channel.*

C Nonstationary Cross-Term Decay

This appendix proves a model nonstationary-phase estimate for cross-term decay.

Lemma C.1 (Localized cross-term decay by nonstationary phase)**.** *Define*

$$Q^{(\rho)}_{\text{cross}}(\tau, L) := 2 \sum_{j=0}^{J} w_j^2 \operatorname{Re} \int_\tau^{\tau+L} w_\rho(u) B_{\rho,j}(u) \overline{B_{\text{rest},j}(u)} e^{i\Phi_j(u)} \, \mathrm{d}u.$$

Assume that, on $[\tau, \tau + L]$,

$$\|w_\rho\|_\infty + \|w'_\rho\|_\infty \le C_w,$$
$$|B_{\rho,j}(u)| + |B'_{\rho,j}(u)| \le C_\rho \tau^2,$$
$$|B_{\text{rest},j}(u)| + |B'_{\text{rest},j}(u)| \le C_{\text{rest}},$$
$$|\Phi'_j(u)| \ge c_\Phi \tau, \qquad |\Phi''_j(u)| \le C_\Phi.$$

Then

$$|Q^{(\rho)}_{\text{cross}}(\tau, L)| \le C_{\text{cross}} \tau L$$

for sufficiently large τ *and all* $L \ge 1$.

Proof. Set

$$g_j(u) := w_\rho(u) B_{\rho,j}(u) \overline{B_{\text{rest},j}(u)}.$$

By the product rule and the assumed bounds,

$$|g_j(u)| + |g'_j(u)| \le C_g \tau^2.$$

Let

$$I_j := \int_\tau^{\tau+L} g_j(u) e^{i\Phi_j(u)} \, \mathrm{d}u.$$

Since Φ'_j is bounded away from zero, write

$$e^{i\Phi_j(u)} = \frac{1}{i\Phi'_j(u)} \frac{\mathrm{d}}{\mathrm{d}u} e^{i\Phi_j(u)}.$$

Integration by parts gives

$$I_j = \left[\frac{g_j(u)}{i\Phi_j'(u)} e^{i\Phi_j(u)} \right]_\tau^{\tau+L} - \int_\tau^{\tau+L} \left(\frac{g_j'(u)}{i\Phi_j'(u)} - \frac{g_j(u)\Phi_j''(u)}{i(\Phi_j'(u))^2} \right) e^{i\Phi_j(u)} \, \mathrm{d}u.$$

Taking absolute values,

$$|I_j| \le 2 \sup \frac{|g_j|}{|\Phi_j'|} + \int_\tau^{\tau+L} \frac{|g_j'|}{|\Phi_j'|} \, \mathrm{d}u + \int_\tau^{\tau+L} \frac{|g_j||\Phi_j''|}{|\Phi_j'|^2} \, \mathrm{d}u.$$

Using the hypotheses,

$$\frac{|g_j|}{|\Phi_j'|} \le C\tau,$$

$$\frac{|g_j'|}{|\Phi_j'|} \le C\tau,$$

and

$$\frac{|g_j||\Phi_j''|}{|\Phi_j'|^2} \le C.$$

Therefore

$$|I_j| \le C\tau + C\tau L + CL \le C'\tau L$$

for $\tau \ge 1$, $L \ge 1$. Summing over finitely many j gives

$$|Q_{\mathrm{cross}}^{(\rho)}(\tau, L)| \le C_{\mathrm{cross}} \tau L.$$

□

Corollary C.2 (Cross term is quartic-subcritical)**.** *If*

$$P_4(\rho; \tau, L) \ge c_4 \tau^4 L,$$

then

$$\frac{|Q_{\mathrm{cross}}^{(\rho)}(\tau, L)|}{P_4(\rho; \tau, L)} \le \frac{C_{\mathrm{cross}}}{c_4 \tau^3} \to 0.$$

Thus

$$Q_{\mathrm{cross}}^{(\rho)} = o(P_4).$$

Proof. Combine

$$|Q_{\mathrm{cross}}^{(\rho)}(\tau, L)| \le C_{\mathrm{cross}} \tau L$$

with

$$P_4(\rho; \tau, L) \ge c_4 \tau^4 L.$$

□

D Gaussian Hammer: Heat-Semigroup Regularization

This appendix records the Gaussian Hammer principle: Gaussian readout converts singular or distributional data into a smooth effective profile and suppresses high frequencies exponentially.

We use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} \, \mathrm{d}x.$$

For $\sigma > 0$, define

$$G_\sigma(x) := \sigma^{-n} e^{-\pi |x|^2/\sigma^2}.$$

Lemma D.1 (Fourier transform of the scaled Gaussian)**.** *With the above convention,*

$$\widehat{G_\sigma}(\xi) = e^{-\pi\sigma^2|\xi|^2}.$$

Proof. The standard identity is

$$\widehat{e^{-\pi|x|^2}}(\xi) = e^{-\pi|\xi|^2}.$$

Since

$$G_\sigma(x) = \sigma^{-n} G_1(x/\sigma),$$

the Fourier scaling rule gives

$$\widehat{G_\sigma}(\xi) = \widehat{G_1}(\sigma\xi) = e^{-\pi\sigma^2|\xi|^2}.$$

□

Theorem D.2 (Gaussian hammer regularizes singular readout)**.** *Let $u \in \mathcal{S}'(\mathbb{R}^n)$ be a tempered distribution and define*

$$u_\sigma := G_\sigma * u.$$

Then:

(1) $u_\sigma \in C^\infty(\mathbb{R}^n)$*;*

(2)

$$\widehat{u_\sigma}(\xi) = e^{-\pi\sigma^2|\xi|^2}\widehat{u}(\xi);$$

(3) *every high-frequency component is exponentially suppressed by*

$$e^{-\pi\sigma^2|\xi|^2};$$

(4) *if $u = \delta_{x_0}$, then*

$$u_\sigma(x) = G_\sigma(x - x_0).$$

Proof. Since $G_\sigma \in \mathcal{S}(\mathbb{R}^n)$ and $u \in \mathcal{S}'(\mathbb{R}^n)$, convolution gives a smooth function. For every multiindex α,

$$\partial^\alpha u_\sigma = \partial^\alpha(G_\sigma * u) = (\partial^\alpha G_\sigma) * u.$$

The right-hand side is well-defined and continuous.

The Fourier convolution identity gives

$$\widehat{u_\sigma} = \widehat{G_\sigma}\widehat{u} = e^{-\pi\sigma^2|\xi|^2}\widehat{u}.$$

The exponential factor suppresses high frequencies.

Finally,

$$G_\sigma * \delta_{x_0} = G_\sigma(\cdot - x_0),$$

so

$$u_\sigma(x) = G_\sigma(x - x_0).$$

□

Corollary D.3 (Gaussian suppression beats every polynomial)**.** *For every $N > 0$, there exists $C_{N,\sigma} > 0$ such that*

$$e^{-\pi\sigma^2|\xi|^2} \le C_{N,\sigma}(1 + |\xi|)^{-N}.$$

Proof. Let $r = |\xi|$. The function

$$r \mapsto e^{-\pi\sigma^2 r^2}(1 + r)^N$$

is continuous on $[0, \infty)$ and tends to 0 as $r \to \infty$. Hence it is bounded. □

Proposition D.4 (Gaussian readout as heat flow)**.** *Let Δ be the Laplacian on $\mathbb{R}^n$. Then*

$$G_\sigma * u = e^{t\Delta}u, \qquad t = \frac{\sigma^2}{4\pi}.$$

Proof. The Fourier transform of the Laplacian is

$$\widehat{\Delta u}(\xi) = -4\pi^2|\xi|^2\widehat{u}(\xi).$$

Thus

$$\widehat{e^{t\Delta}u}(\xi) = e^{-4\pi^2 t|\xi|^2}\widehat{u}(\xi).$$

Choosing

$$t = \frac{\sigma^2}{4\pi}$$

gives

$$e^{-4\pi^2 t|\xi|^2} = e^{-\pi\sigma^2|\xi|^2}.$$

This is exactly the Fourier multiplier for convolution by G_σ. □

Remark D.5 (Gravity-facing interpretation)**.** *This does not prove removal of physical singularities in general relativity. It proves the analytic readout statement:*

$$\textit{singular raw source} \quad \longmapsto \quad \textit{smooth Gaussian effective profile.}$$

Singularity wants a spike. Gaussian hammer returns a scale.

E Source–Love Channel 2.0: A Separate Meta-Model

This appendix is not used as an input to the RH contradiction. It records a separate operator model for the general principle that stable structure arises from the interaction of transport, controlled perturbation, and contraction.

The Source–Love Channel should therefore be read as a conceptual companion to the localized bridge framework, not as a proof component for RH.

Remark E.1 (Role of the Source–Love Channel)**.** *The Source–Love Channel is not used in the proof of the fixed-channel contradiction theorem. It is included only as a companion model for the general principle that stable readout can arise from transport, perturbation, and contraction.*

Thus none of the RH implications in the main body depend on the Source–Love Channel.

Let

$$\mathcal{H} \simeq \mathbb{C}^d$$

and let $\mathcal{D}(\mathcal{H})$ denote the set of density operators on $\mathcal{H}$. Fix:

- a unitary operator $U : \mathcal{H} \to \mathcal{H}$;
- a target state $\rho_\infty \in \mathcal{D}(\mathcal{H})$;
- a source state $\rho_F \in \mathcal{D}(\mathcal{H})$;
- parameters
$$\lambda \in (0, 1], \qquad \gamma \in [0, 1).$$

Define the Source–Love Channel by

$$\mathcal{T}(\rho) = (1-\lambda)(1-\gamma)U\rho U^\dagger + (1-\lambda)\gamma U\rho_F U^\dagger + \lambda\rho_\infty.$$

The three terms encode:

$$\text{transport} + \text{source perturbation} + \text{stabilizing contraction}.$$

Theorem E.2 (Source–Love contraction)**.** *For all $\rho, \sigma \in \mathcal{D}(\mathcal{H})$,*

$$\|\mathcal{T}(\rho) - \mathcal{T}(\sigma)\|_1 = (1-\lambda)(1-\gamma)\|\rho - \sigma\|_1.$$

Hence $\mathcal{T}$ is a strict contraction whenever

$$q := (1-\lambda)(1-\gamma) < 1.$$

Proof. Subtracting gives

$$\mathcal{T}(\rho) - \mathcal{T}(\sigma) = (1-\lambda)(1-\gamma)U(\rho - \sigma)U^\dagger.$$

The trace norm is unitarily invariant:

$$\|U(\rho - \sigma)U^\dagger\|_1 = \|\rho - \sigma\|_1.$$

Therefore

$$\|\mathcal{T}(\rho) - \mathcal{T}(\sigma)\|_1 = (1-\lambda)(1-\gamma)\|\rho - \sigma\|_1.$$

□

Corollary E.3 (Unique stable attractor)**.** *The map $\mathcal{T}$ has a unique fixed point $\rho_* \in \mathcal{D}(\mathcal{H})$, and*

$$\|\mathcal{T}^n(\rho) - \rho_*\|_1 \le q^n \|\rho - \rho_*\|_1.$$

Proof. This is the Banach fixed-point theorem applied to the complete metric space

$$(\mathcal{D}(\mathcal{H}), \|\cdot\|_1).$$

□

E.1 Fixed point via Neumann series

Let

$$c := (1-\lambda)(1-\gamma),$$

and define

$$b := (1-\lambda)\gamma U \rho_F U^\dagger + \lambda \rho_\infty.$$

Then the fixed point equation is

$$\rho_* = c\, U \rho_* U^\dagger + b.$$

Equivalently,

$$(I - c\,\mathrm{Ad}_U)\rho_* = b,$$

where

$$\mathrm{Ad}_U(X) := UXU^\dagger.$$

Proposition E.4 (Neumann-series representation)**.** *The unique fixed point satisfies*

$$\rho_* = \sum_{n=0}^{\infty} c^n\, \mathrm{Ad}_U^n(b).$$

Proof. Since

$$\|\,\mathrm{Ad}_U\,\|_{1\to 1} = 1$$

and $0 \le c < 1$, the operator $I - c\,\mathrm{Ad}_U$ is invertible by the Neumann series:

$$(I - c\,\mathrm{Ad}_U)^{-1} = \sum_{n=0}^{\infty} c^n\, \mathrm{Ad}_U^n\,.$$

Applying this to b gives the representation. □

E.2 Progress as reduction of epistemic debt

Define the epistemic debt

$$D(\rho) := \|\rho - \rho_*\|_1.$$

Then

$$D(\mathcal{T}\rho) = qD(\rho).$$

Definition E.5 (Progress functional)**.** *Define*

$$\mathcal{P}(\rho) := D(\rho) - D(\mathcal{T}\rho).$$

Hence

$$\mathcal{P}(\rho) = (1-q)D(\rho).$$

Theorem E.6 (Progress is positive away from the attractor)**.** *If $\rho \neq \rho_*$, then*

$$\mathcal{P}(\rho) > 0.$$

If $\rho = \rho_*$, then

$$\mathcal{P}(\rho) = 0.$$

Proof. Since $1 - q > 0$,

$$\mathcal{P}(\rho) = (1-q)\|\rho - \rho_*\|_1.$$

This is positive exactly when $\rho \neq \rho_*$. □

E.3 Continuous-time limit

Let

$$\lambda = \kappa\Delta t, \qquad \gamma = \mu\Delta t, \qquad U = e^{-iH\Delta t}.$$

Then

$$U\rho U^\dagger = \rho - i\Delta t[H,\rho] + o(\Delta t).$$

Proposition E.7 (Continuous-time Lindblad-type limit)**.** *In the limit $\Delta t \to 0$, the Source–Love dynamics converge formally to*

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -i[H,\rho] + \kappa(\rho_\infty - \rho) + \mu(\rho_F - \rho).$$

Proof. Substitute

$$\lambda = \kappa\Delta t, \qquad \gamma = \mu\Delta t, \qquad U = e^{-iH\Delta t}$$

into $\mathcal{T}$. Keeping only first-order terms gives

$$\mathcal{T}(\rho) - \rho = \Delta t\,(-i[H,\rho] + \kappa(\rho_\infty - \rho) + \mu(\rho_F - \rho)) + o(\Delta t).$$

Dividing by Δt and taking $\Delta t \to 0$ gives the stated generator. □

Progress is not doing everything possible.

Progress is converting epistemic debt into stable readout.

Love is not sentiment; it is the contraction that lets exploration survive without exploding.